

Statistical modeling

Module 05

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Bachelor, GSEM

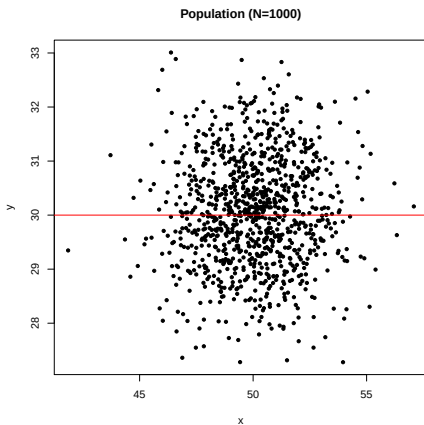
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Inferential problem

Reminder: inference is about validating/rejecting a hypothesis about a population when we can only observe a sample drawn at random from this population.

Inferential problem

Example: Below the population of times to achieve a task (y) and the volume of the music listen when this task was made (x). These are simulated data. There are independent: y does not vary with x .

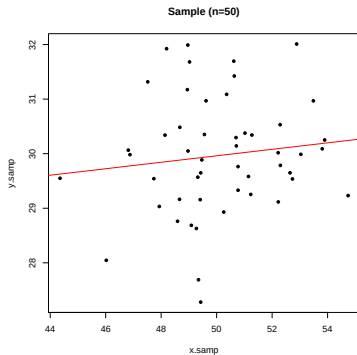


Inferential problem

Suppose that, in a study, the researcher wants to see the effect of the volume x on the time of the task y . The researcher however does not have access to $N = 1000$ observation (the population) but only to a sample of it ($n = 50$), selected at random. The results are:

Model equation:

$$y = 27 + 0.06x + e$$



Inferential problem

From the research point of view (i.e. "is x influencing y , in the population?"), the regression obtained from the sample does not help:

- ▶ The coefficient 0.06 is non-zero, thus x has an impact on y
- ▶ The coefficient is however very small, thus x may not have an real impact.

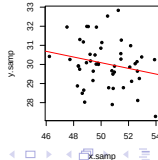
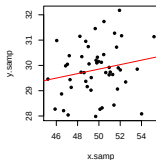
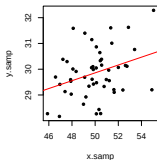
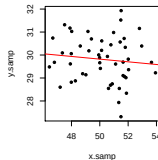
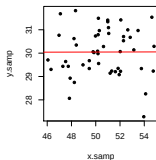
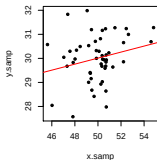
Of course the coefficient cannot be exactly 0. This is just impossible. It is non-null at a given precision level.

In this context, the statistical inference answers the question

"When is a coefficient small enough to conclude that the corresponding variable x has no influence on y ?"

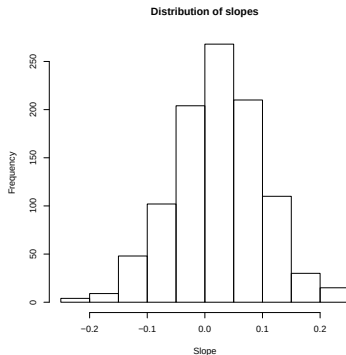
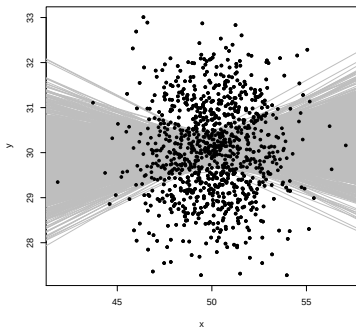
Inferential problem

When the question is related to the population (i.e. that cannot be completely observed), the question should not be "what happened with the sample?" but "what could happen with the sample?". To illustrate, here are 6 possible samples from the population



Inferential problem

If we look at (almost...) all the possible samples that could have been extracted from this population, then we could have



Inferential problem

The distribution of the possible slopes looks like the normal distribution. In fact, knowing that there is no link between x and y in the population, this normality could have been anticipated from the mathematical theory.

Without going into the details of this theory, we will use it to make inference on our regression to draw conclusions not only at the level of the sample but also at the level of the population.

Inferential problem

Mainly two types of problems with regression

- ▶ Prediction: use a sample $\{y_i, x_i\}_{i=1}^n$ already observed to build a trustworthy prediction of y for a new x
- ▶ Research: discover whether a link between y and one or more variables of interest x exists.

If we knew the population model,

$$y = b_0 + b_1x_1 + \cdots + b_px_p + e,$$

then we would easily answer these questions. However, we don't know it. It is only estimated from a sample.

Assuming that the sample is randomly extracted from the population of interest, then one may use statistical theory to answer these questions.

Model significance: F-test

Suppose that, in the population, there is no link at all between y and x . The true population model would thus be $y = b_0 + e$. This is translated into the statistical null hypothesis:

$$H_0 : b_1 = \dots = b_p = 0$$

As illustrated before, using the sample will NEVER give exactly $\hat{b}_1 = \dots = \hat{b}_p = 0$ exactly. So

What could happen if H_0 is true?¹

¹Remember, this is the good question

Model significance: F-test

If H_0 is true then, the mathematical theory tells us that the statistic

$$F = \frac{MS_R}{MS_e} = \frac{SS_R/p}{SS_e/(n - p - 1)}$$

is distributed according to the Fisher distribution

$$F_{df_1, df_2}$$

where the degrees of freedom are

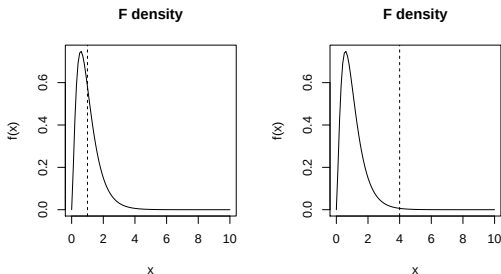
$$df_1 = k, \quad df_2 = n - p - 1.$$

Above, MS stands for "Mean Squares".

Interpretation: if we could repeat the sampling (and the regression fit with the new sample) a very large number of times, computing F for each of these sample, and looking at their distribution, then we would see a Fisher distribution.

Model significance: F-test

Thus, to test if H_0 is supported by the data, the observed value F is compared to the distribution F_{df_1, df_2} , which represents H_0 .



- ▶ Left: the observed F is 1.1. It could come from the F distribution. H_0 and the data are compatibility: do not reject H_0 .
- ▶ Right: the observed F is 4. It is very unlikely F distributed. H_0 and the data are incompatibility: reject H_0 .

Model significance: F-test

For which value of F should we switch from "Do not reject H_0 " to "Reject H_0 "?

The area to the right of F is the p -value

$$p = 1 - F_{df_1, df_2}(F)$$

When this p -value is small (F is large compare to the Fisher distribution) then reject H_0 . The limit of the decision is α , the significance level of the test. Often², $\alpha = 5\%$.

Equivalently, reject H_0 when

- ▶ $p\text{-value} < \alpha$,
- ▶ $F > F_{df_1, df_2; 1-\alpha}$.

²and in this course except mentioned

Model significance: F-test

Procedure

- ▶ Compute $F = \frac{MS_R}{MS_e}$,
- ▶ Compute $F_{df_1, df_2; 1-\alpha}$ (read in table or use R),
- ▶ Reject $H_0 : b_1 = \dots = b_p = 0$ if either

$$F > F_{df_1, df_2; 1-\alpha}$$

or

$$p = 1 - F_{df_1, df_2}(F) < \alpha.$$

In R,

- ▶ $F_{df_1, df_2; 1-a} = \text{qf}(1-a, \text{df1}, \text{df2})$
- ▶ $p = 1 - \text{pf}(F, \text{df1}, \text{df2})$

Model significance: F-test

Example: For the burn out study, restricted to males,

```
> fitM <- lm(BurnOut~Social+I(Social^2), subset=Sex=="M", data=ashley)
> summary(fitM)
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	149.59384	187.90193	0.796	0.4521
Social	-4.40025	7.80315	-0.564	0.5904
I(Social^2)	0.15926	0.06968	2.286	0.0562 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 79.47 on 7 degrees of freedom

Multiple R-squared: 0.951, Adjusted R-squared: 0.937

F-statistic: 67.93 on 2 and 7 DF, p-value: 2.604e-05

$df_1 = 2$, $df_2 = 7$, $F = 67.93$. The model is significant at 5% because $F > 4.74 = F_{2,7;0.95}$ (not shown by R). The corresponding p -value is 2.604×10^{-5} .

Model significance: in practice

Whenever a regression model is built, look at the global F -test and its p -value.

- ▶ If H_0 is rejected, use the model as usual.
- ▶ If H_0 is not rejected (at 5%), the model is declared non-significant. The observed link between y and the covariates x is probably due to chance (the sampling).

The F statistics and the R^2 's are intimately linked. Only the F statistic can be used to test the significance of the model.

Even with a small R^2 , a significant model can be used. It is "better than nothing".

ANOVA: ANalysis Of VAriance

The ANOVA table summarizes the F -test calculations. It is often used independently of the regression concept and taken as a theory in itself.

Source	df	SS	MS	F	p
Regression	$df_1 = p$	SS_R	$MS_R = SS_R/df_1$	$F = MS_R/MS_e$	$1 - F_{df_1, df_2}(F)$
Residual	$df_2 = n - p - 1$	SS_e	$MS_e = SS_e/df_2$		
Total	$n - 1$	SS_T			

ANOVA and F-test

Example: Remember the real estate example (pricing of houses). It gives³

Source	df	SS	MS	F	p
Regression	5	$1.04e^9$	$2.1e^8$	277.8	≤ 0.001
Residual	313	$0.23e^9$	$7.5e^5$		
Total	318	$1.27e^9$			

For a test at 5%, use equivalently

- ▶ $277.8 > 3.08 = F_{5,313;0.95}$
- ▶ $p < 0.05$

Therefore reject H_0 , the model is significant (at 5%).

³" e^k " stands for " $\times 10^k$ "

Comparison of two nested models

The F -test is a global test for all the covariates at once. Often a subset of the covariates needs to be tested in order to answer a research question. The F -test can also be used in this case after a small adaptation.

Example: Burn out study: Is there a difference between males and females? To answer, one has to compare models with and without the factor Sex. Consider the two models below:

$$M_1 : y = b_0 + b_1x_1 + b_2x_1^2 + (c_0 + c_1x_1 + c_2x_1^2)\delta_M(x_2) + e$$

versus

$$M_0 : y = b_0 + b_1x_1 + b_2x_1^2 + e$$

where x_1 is the socialization index and x_2 the sex.

Comparison of two nested models

The research hypothesis:

"There is no difference between males and females in the population"

is equivalent to the statistical hypothesis

$$H_0 : c_0 = c_1 = c_2 = 0$$

or also, that the two models are equal: $M_0 = M_1$. Overall, it means that we should not need covariate Sex in the model.

This can be tested using the difference in the sum of squares of each model.

Comparison of two nested models

From the predictions $\hat{y}_{i0}$ and $\hat{y}_{i1}$ of models M_0 and M_1 , the sum of squares of each models are:

$$SS_0 = \sum_{i=1}^n (y_i - \hat{y}_{i0})^2, \quad SS_1 = \sum_{i=1}^n (y_i - \hat{y}_{i1})^2$$

Obviously, $SS_1 \leq SS_0$. Indeed, the full model M_1 is better than M_0 because M_1 has more parameters.

The question is whether this extra complexity (more parameters) is significant.

Comparison of two nested models

Under the model assumptions, it can be shown that, **if H_0 is true**, then the statistics

$$F = \frac{(SS_0 - SS_1)/(df_1 - df_0)}{SS_1/(n - p - 1)} \sim F_{df_1 - df_0, n - p - 1}$$

where

- ▶ n is the sample size,
- ▶ $df_1 = p + 1$ is the number of parameters in model M_1 ,
- ▶ df_0 is the number of parameters in model M_0 .

The test decision is thus to reject H_0 (at level α) if either

- ▶ $F > F_{df_1 - df_0, n - p - 1; 1 - \alpha}$
- ▶ $p\text{-value} = 1 - F_{df_1 - df_0, n - p - 1}(F) < \alpha$

Comparison of two nested models

Example: in the burn out case, the predictions are

$$\begin{aligned}\hat{y}_{i0} &= \hat{b}_{00} + \hat{b}_{01}x_{i1} + \hat{b}_{02}x_{i1}^2 \\ \hat{y}_{i1} &= \hat{b}_{10} + \hat{b}_{11}x_{i1} + \hat{b}_{12}x_{i1}^2 + (\hat{c}_{10} + \hat{c}_{11}x_{i1} + \hat{c}_{12}x_{i1}^2)\delta_M(x_{i2}).\end{aligned}$$

The sum of squares are

$$SS_0 = 361115, SS_1 = 127317, F = 8.57$$

$$n = 20, p = 5, df_1 = p + 1 = 6, df_0 = 3.$$

Thus $F > F_{3,14;0.95} = 3.34$ and H_0 is rejected. Equivalently,

$$p = 1 - F_{3,14}(8.57) = 0.001769 < 0.05.$$

Comparison of two nested models: example

With R,

```
> fit1 <- lm(BurnOut~(Social+I(Social^2))*Sex, data=ashley)
> fit0 <- lm(BurnOut~Social+I(Social^2), data=ashley)
> anova(fit0, fit1)
```

Analysis of Variance Table

```
Model 1: BurnOut ~ Social + I(Social^2)
Model 2: BurnOut ~ (Social + I(Social^2)) * Sex
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	17	361115				
2	14	127317	3	233797	8.5696	0.001769 **

At 5%, the impact of the socialization index on the Burn Out index is significantly different between Males and Females.

ANOVA in research

Often researchers do not (want to) use regression to perform a test. They are interested into testing the significance of an effect directly. This is why automatic ANOVA procedures have been developed in many statistical computer programs. Lots of confusion can arise from this and we now try to give a clear presentation.

- ▶ When ANOVA is applied to a model, several tests are made.
- ▶ Usually, there are 3 types of ANOVA, sadly called "Sum of type I", "Sum of type II", "Sum of type III".
- ▶ To distinguish which one to use and what is done, the important thing is to understand what is tested (rather than what type of sum is computed).
- ▶ In practice, the most used are type II and type III.

ANOVA: Sum of Type III

The ANOVA of type III is testing if the model with or without the term, keeping all the other terms.

Thus,

- ▶ On row `Social:Sex`, the models are

`BurnOut ~ Social + Sex + Social:Sex` vs `BurnOut ~ Social + Sex`

- ▶ On row `Social`, the models are

`BurnOut ~ Social + Sex + Social:Sex` vs `BurnOut ~ Sex + Social:Sex`

- ▶ On row `Sex`, the models are

`BurnOut ~ Social + Sex + Social:Sex` vs `BurnOut ~ Social + Social:Sex`

ANOVA: Sum of Type II

With Anova of type II

```
> Anova(fit, type=2)
Anova Table (Type II tests)

Response: BurnOut
      Sum Sq Df F value    Pr(>F)
Social   874399  1  74.072 2.122e-07 ***
Sex      135742  1  11.499  0.003730 **
Social:Sex 136134  1  11.532  0.003692 **
Residuals 188875 16
[...]
```

Conclusion (test at 5%): keeping all the other terms

- ▶ The interaction is different from 0.
- ▶ That's it...

We could also conclude that, **without the interaction**,

- ▶ The main effect Sex is different from 0,
- ▶ The main effect Social is different from 0.

ANOVA: Sum of Type II

The ANOVA of type II is testing if the model with or without the term, not keeping the interaction terms.

Thus,

- ▶ On row `Social:Sex`, the models are⁵

BurnOut $\sim$ Social + Sex + Social:Sex vs BurnOut $\sim$ Social + Sex

- ▶ On row Social, the models are

BurnOut ~ Social + Sex vs BurnOut ~ Sex

- ▶ On row Sex, the models are

BurnOut ~ Social + Sex vs BurnOut ~ Social

⁵same as type III

ANOVA: Sum of Type II or of type III

If interactions are significant, then type II is often misinterpreted.
If interactions are not significant, type III can only be used to infer the interactions, and in general type II should be preferred.

Example:

```
> fit2 <- lm(log(Price)~Certif*Clarity, data=diamond2)
```

```
> Anova(fit2, type=2)  
Anova Table (Type II tests)
```

Response: log(Price)

	Sum Sq	Df	F value	Pr(>F)
Certif	24.312	1	49.423	1.412e-11 ***
Clarity	20.755	4	10.548	5.258e-08 ***
Certif:Clarity	3.530	4	1.794	0.13
Residuals	146.590	298		

```
> Anova(fit2, type=3)  
Anova Table (Type III tests)
```

Response: log(Price)

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	305.632	1	621.3135	< 2.2e-16 ***
Certif	6.792	1	13.8067	0.0002419 ***
Clarity	0.597	4	0.3035	0.8755167
Certif:Clarity	3.530	4	1.7940	0.1299562
Residuals	146.590	298		

What can be concluded here?

ANOVA: Sum of Type I

The ANOVA of type I is incremental. Read from bottom to top, it test the model with or with all the terms below.

Thus,

- ▶ On row `Social:Sex`, the models are⁶

`BurnOut ~ Social + Sex + Social:Sex` vs `BurnOut ~ Social + Sex`

- ▶ On row `Sex`, the models are⁷

`BurnOut ~ Social + Sex` vs `BurnOut ~ Social`

- ▶ On row `Social`, the models are

`BurnOut ~ Social` vs `BurnOut ~ Sex`

⁶same as type II and III

⁷same as type II

ANOVA: Sum of Type I

```
> anova(fit)
Analysis of Variance Table

Response: BurnOut
      Df Sum Sq Mean Sq F value    Pr(>F)
Social    1  932504   932504   78.994 1.38e-07 ***
Sex        1  135742   135742   11.499 0.003730 **
Social:Sex  1  136134   136134   11.532 0.003692 **
Residuals 16 188875    11805
[...]
```

Conclusion is the same as for type II in this case.

ANOVA of type I is rarely used in practice. Except for balanced design*, it depends on the order of the terms in the model. It is mainly for large models when variable selection is of interest.

It is the result from R function `anova(mod)`. Prefer function `Anova(mod, type 2 or 3)` from package `car`.

ANOVA and F-test: conclusions

In general, prefer a well controlled comparison of models with `anova(mod1, mod2)`.

In practice, declaring not significant a variable can be

- ▶ The purpose of a research study (see later).
- ▶ The occasion to save resources for prediction (e.g. simplify data base in marketing applications, avoid having to track data, etc.).